

Wenshu Zhao

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EDUCATION

University of Washington, Seattle

Sep.2024 - Jun.2026(Expected)

Master of Science in Information Science (Specialization: Data Science & Artificial Intelligence)

GPA: 3.9/4.0

China Pharmaceutical University, Nanjing

Sep.2020 - Jun.2024

Bachelor of Science in Information Management and Information Systems

GPA: 3.7/4.0

SKILLS

Programming Languages: Java, Python, JavaScript, SQL, C++, TypeScript, HTML/CSS

Frameworks & Tools: Spring Boot, React, Vue, Hadoop, Apache Spark, TensorFlow, PyTorch, Git, Expo, AuthN/AuthZ

Databases & Caching: MySQL, SQL Server, NoSQL, Redis

Cloud & DevOps: AWS, Azure, Docker, Linux, CI/CD, Nginx

Concepts: Distributed Systems, Microservices, System Design, Data Structures, Software Development Life Cycle

PROFESSIONAL EXPERIENCE

Full-Stack Engineer Intern - IpserLab

Jun.2025 - Dec.2025

- Developed backend modules for a municipal permit application using an internal ORM framework and **Java**, modeling 15+ XML-defined entities across User, Application, Payment, and Document domains.
- Designed a schema-driven form system in **React Native** with **TypeScript** that supports 6-stage workflows with 40+ dynamic fields, reducing code duplication by **60%** through reusable components and custom hooks.
- Built a multi-currency payment engine supporting **USD** and **ZIG** currencies with dynamic fee computation based on service type, density classification, and cluster count, integrating 15% VAT calculations with real-time updates.
- Implemented secure authentication system with email validation, password strength requirements, 2FA, and session token management with user-scoped data isolation, preventing duplicate registrations and ensuring data privacy.

Full-Stack Engineer Intern - Legend Biotech - IT Department

Feb.2023 - Apr.2023

- Streamlined daily check-in workflows for **500+** employees and improved employee attendance tracking efficiency by 40% across mobile platforms by developing a cross-platform check-in application, using **Java** and **JavaScript**.
- Built patient data management modules for a CAR-T treatment tracking system using **Spring Boot** and improved frontend user experience with **React**, enabling real-time treatment schedule updates and progress tracking for 200+ patients, in collaboration with clinical staff and full-stack engineers.

PROJECT EXPERIENCE

Seattle Source Ranker

Sep.2025 - Dec.2025

- Built a distributed ingestion system with **Celery & Redis** to collect and validate **400K+** repositories from **28K** Seattle developers, achieving **8.1×** throughput improvement and **99.98%** ingestion reliability under strict rate limits.
- Architected a fault-tolerant API orchestration layer combining **GraphQL** and **REST fallback** strategies to correct inconsistent watcher metrics, eliminating **~2%** invalid repositories and ensuring ranking data integrity at scale.
- Implemented a rate-limit-aware token rotation system (6 tokens $\times$ 5K req/hr) with adaptive backoff and idempotent retries, reducing rate-limit failures by 92% and enabling full dataset rebuilds within 60-90 minutes.
- Designed a ranking pipeline integrating PyPI's **700K+** package index and global Top-15K rankings, identifying 1,075 Seattle PyPI packages and surfacing 28 globally significant projects via offline matching with 100% precision.
- Automated daily dataset updates and static deployment via GitHub Actions (**Bash** scripts), generating **9K+** **optimized JSON datasets** consumed by a React interface with real-time search and sub-50ms page load times.

Virtual Character Letter Platform

Nov.2024 - Apr.2025

- Built a Python **FastAPI** service for LLM-powered letter generation with dynamic character-specific prompts, multi-language support and multi-turn context, integrating **GPT** and **Doubao** via OpenAI-compatible APIs.
- Refactored LLM integration using the **Strategy Pattern**, enabling zero-change backend extension; added exponential backoff retry (1s $\rightarrow$ 2s $\rightarrow$ 4s) with 4xx short-circuit to eliminate redundant retries on auth failures.
- Architected a two-layer async communication system to decouple Java-Python interactions: **FastAPI BackgroundTasks** for non-blocking HTTP responses; and **RabbitMQ** as a message broker for reliable cross-service task delivery with persistent queuing and automatic retry on failure.
- Reduced response latency and supported flexible data modeling by implementing **Redis** caching and a multi-database architecture, using **MongoDB** for unstructured letter content and **MySQL** for structured user and role data.
- Ensured reliable deployment and operational stability by containerizing services with **Docker**, deploying to **Amazon EC2**, configuring **Nginx** as a reverse proxy, and setting up **AWS CloudWatch** for monitoring and logging.